

```
In [2]: import pycaret
from pycaret.classification import *
from pandas import read_csv, concat

def tolower(s) :
    return s.lower()

url = 'full-list-1974-2023.csv'
df = read_csv(url, usecols=['sex', 'FirstForename'], converters={'FirstForename':to

df[['n1', 'n2', 'n3', 'n4', 'n5', 'n6', 'n7', 'n8', 'n9', 'n10']] = \
    df['FirstForename'].str.extract( \
        '(?)(?)(?)(?)(?)(?)(?)(?)(?)(?)', expand = True)
```

```
In [3]: df = df.drop('FirstForename', axis=1)
df = df.drop_duplicates()
df = df.reset_index(drop=True)
print(df.shape)
print(df.head())
print(df.tail())
```

```
(75810, 11)
   sex n1 n2 n3 n4 n5 n6 n7 n8 n9 n10
0    B  a  a  m  i  r
1    B  a  a  r  o  n
2    B  a  b  d  u  l
3    B  a  b  d  u  l  l  a  h  i
4    B  a  b  d  u  l  r  a  z  a  k
      sex n1 n2 n3 n4 n5 n6 n7 n8 n9 n10
75805   G  z  o  e  -  m  o  n  i  q  u
75806   G  z  o  r  a  y  a
75807   G  z  o  r  y  a  n  a
75808   G  z  u  m  o  r  r  u  d
75809   G  z  y  o  n  -  j  a  i
```

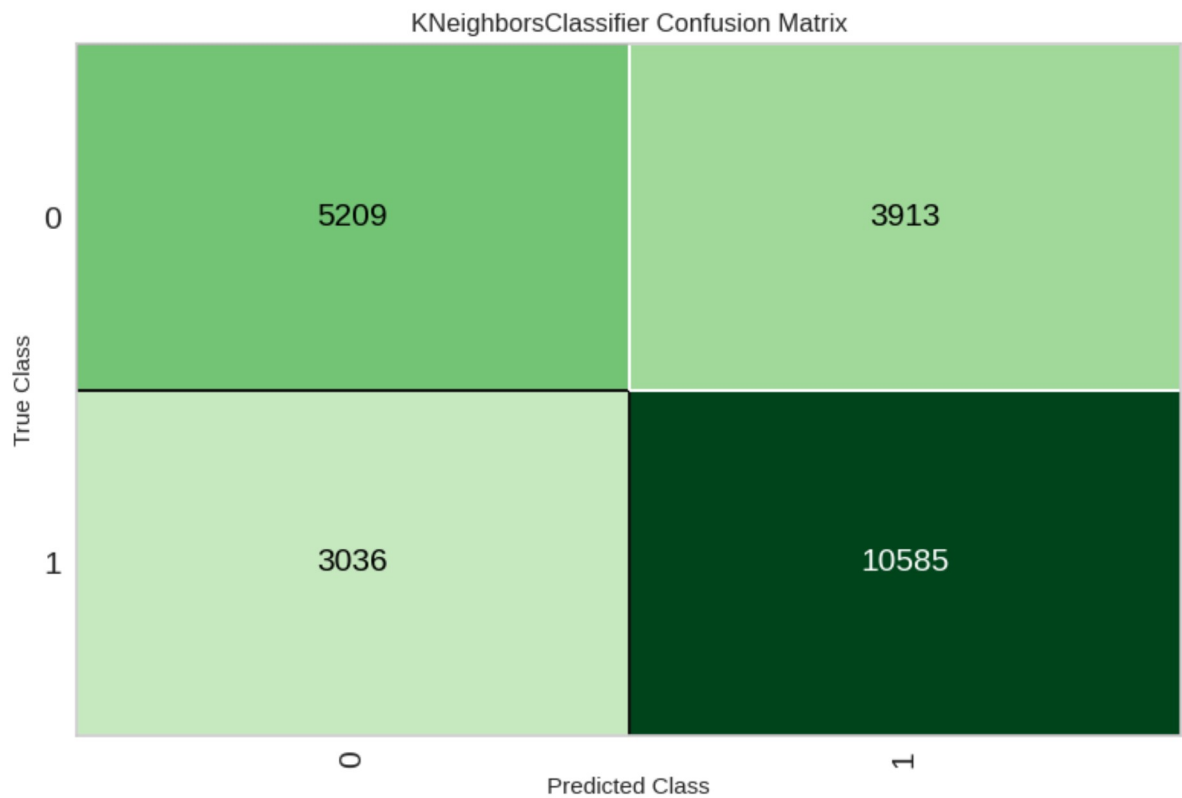
```
In [4]: grid = setup(data=df, target='sex', session_id=42)
```

	Description	Value
0	Session id	42
1	Target	sex
2	Target type	Binary
3	Target mapping	B: 0, G: 1
4	Original data shape	(75810, 11)
5	Transformed data shape	(75810, 11)
6	Transformed train set shape	(53067, 11)
7	Transformed test set shape	(22743, 11)
8	Categorical features	10
9	Preprocess	True
10	Imputation type	simple
11	Numeric imputation	mean
12	Categorical imputation	mode
13	Maximum one-hot encoding	25
14	Encoding method	None
15	Fold Generator	StratifiedKFold
16	Fold Number	10
17	CPU Jobs	-1
18	Use GPU	False
19	Log Experiment	False
20	Experiment Name	clf-default-name
21	USI	0684

In [6]: `best = compare_models(sort='Precision')`

	Model	Accuracy	AUC	Recall	Prec.	F1	Kappa	MCC	TT (Sec)
knn	K Neighbors Classifier	0.7004	0.7572	0.7872	0.7325	0.7588	0.3646	0.3663	1.2300
nb	Naive Bayes	0.6447	0.6844	0.6637	0.7209	0.6910	0.2746	0.2760	0.2390
qda	Quadratic Discriminant Analysis	0.6553	0.6983	0.7102	0.7131	0.7116	0.2833	0.2833	0.2470
lda	Linear Discriminant Analysis	0.6661	0.7153	0.8034	0.6901	0.7424	0.2757	0.2823	0.2820
lr	Logistic Regression	0.6657	0.7150	0.8047	0.6892	0.7425	0.2741	0.2809	0.5980
ridge	Ridge Classifier	0.6657	0.0000	0.8071	0.6884	0.7430	0.2732	0.2804	0.2350
svm	SVM - Linear Kernel	0.6606	0.0000	0.8443	0.6749	0.7483	0.2447	0.2640	0.2530
et	Extra Trees Classifier	0.6473	0.7480	0.9476	0.6385	0.7629	0.1662	0.2283	0.9900
dt	Decision Tree Classifier	0.5971	0.5032	0.9779	0.6005	0.7440	0.0077	0.0249	0.2410
lightgbm	Light Gradient Boosting Machine	0.5980	0.4856	0.9839	0.6003	0.7456	0.0069	0.0250	0.3290
gbc	Gradient Boosting Classifier	0.5993	0.5522	0.9922	0.6001	0.7478	0.0057	0.0259	1.2110
rf	Random Forest Classifier	0.5981	0.5133	0.9892	0.5997	0.7467	0.0039	0.0267	0.7880
ada	Ada Boost Classifier	0.5977	0.6061	0.9886	0.5995	0.7464	0.0032	0.0157	0.5630
dummy	Dummy Classifier	0.5989	0.5000	1.0000	0.5989	0.7491	0.0000	0.0000	0.2460

```
In [7]: plot_model(best, 'confusion_matrix')
```



```
In [10]: besttuned = tune_model(best)
bestthreshold = optimize_threshold(besttuned)
```

	Accuracy	AUC	Recall	Prec.	F1	Kappa	MCC
Fold							
0	0.7304	0.7960	0.7671	0.7792	0.7731	0.4409	0.4410
1	0.7360	0.8053	0.7728	0.7834	0.7781	0.4524	0.4524
2	0.7304	0.7987	0.7703	0.7774	0.7738	0.4400	0.4401
3	0.7239	0.7892	0.7668	0.7710	0.7689	0.4262	0.4262
4	0.7305	0.8021	0.7706	0.7775	0.7740	0.4404	0.4404
5	0.7177	0.7834	0.7638	0.7647	0.7642	0.4126	0.4126
6	0.7198	0.7871	0.7666	0.7659	0.7662	0.4166	0.4166
7	0.7292	0.7969	0.7697	0.7763	0.7729	0.4375	0.4375
8	0.7262	0.7968	0.7631	0.7760	0.7695	0.4323	0.4324
9	0.7337	0.8001	0.7788	0.7771	0.7779	0.4454	0.4454
Mean	0.7278	0.7956	0.7690	0.7748	0.7719	0.4344	0.4345
Std	0.0055	0.0066	0.0044	0.0056	0.0044	0.0119	0.0120

Fitting 10 folds for each of 10 candidates, totalling 100 fits

```
-----
KeyboardInterrupt                                Traceback (most recent call last)
File /opt/conda/envs/pycaret/lib/python3.8/site-packages/joblib/parallel.py:975, in
Parallel.retrieve(self)
    974 if getattr(self._backend, 'supports_timeout', False):
--> 975     self._output.extend(job.get(timeout=self.timeout))
    976 else:

File /opt/conda/envs/pycaret/lib/python3.8/site-packages/joblib/_parallel_backend
s.py:567, in LokyBackend.wrap_future_result(future, timeout)
    566 try:
--> 567     return future.result(timeout=timeout)
    568 except CfTimeoutError as e:

File /opt/conda/envs/pycaret/lib/python3.8/concurrent/futures/_base.py:439, in Futu
re.result(self, timeout)
    437 return self.__get_result()
--> 439 self._condition.wait(timeout)
    441 if self._state in [CANCELLED, CANCELLED_AND_NOTIFIED]:

File /opt/conda/envs/pycaret/lib/python3.8/threading.py:302, in Condition.wait(sel
f, timeout)
    301 if timeout is None:
--> 302     waiter.acquire()
    303     gotit = True
```

KeyboardInterrupt:

During handling of the above exception, another exception occurred:

```
KeyboardInterrupt                                Traceback (most recent call last)
Cell In[10], line 2
      1 besttuned = tune_model(best)
----> 2 bestthreshold = optimize_threshold(besttuned)

File /opt/conda/envs/pycaret/lib/python3.8/site-packages/pycaret/utils/generic.py:9
65, in check_if_global_is_not_none.<locals>.decorator.<locals>.wrapper(*args, **kwa
rgs)
    963 if globals_d[name] is None:
    964     raise ValueError(message)
--> 965 return func(*args, **kwargs)

File /opt/conda/envs/pycaret/lib/python3.8/site-packages/pycaret/classification/fun
ctional.py:2115, in optimize_threshold(estimator, optimize, return_data, plot_kwarg
s, **shgo_kwargs)
    2057 @check_if_global_is_not_none(globals(), _CURRENT_EXPERIMENT_DECORATOR_DICT)
    2058 def optimize_threshold(
    2059     estimator,
    (...)
    2063     **shgo_kwargs,
    2064 ):
    2065     """
    2066     This function optimizes probability threshold for a trained classifier.
It
    2068     uses the SHGO optimizer from ``scipy`` to optimize for the given metri
c.
```

```

(...)
2112     - This function does not support multiclass classification problems.
2113     """
-> 2115     return _CURRENT_EXPERIMENT.optimize_threshold(
2116         estimator=estimator,
2117         optimize=optimize,
2118         return_data=return_data,
2119         plot_kwags=plot_kwags,
2120         **shgo_kwags,
2121     )

```

File /opt/conda/envs/pycaret/lib/python3.8/site-packages/pycaret/classification/oo
p.py:2722, in ClassificationExperiment.optimize_threshold(self, estimator, optimize
, return_data, plot_kwags, **shgo_kwags)

```

2719         super().__init__(self.seed)
2721     with patch("numpy.random.RandomState", FixedRandom):
-> 2722         result = shgo(objective, ((0, 1),), **shgo_kwags)
2724     message = (
2725         "optimization loop finished successfully. "
2726         f"Best threshold: {result.x[0]} with {optimize}={result.fun*direction}"
2727     )
2728     print(message)

```

File /opt/conda/envs/pycaret/lib/python3.8/site-packages/scipy/optimize/_shgo.py:42
6, in shgo(func, bounds, args, constraints, n, iters, callback, minimizer_kwags, o
ptions, sampling_method)

```

420     shc = SHGO(func, bounds, args=args, constraints=constraints, n=n,
421                 iters=iters, callback=callback,
422                 minimizer_kwags=minimizer_kwags,
423                 options=options, sampling_method=sampling_method)
425     # Run the algorithm, process results and test success
-> 426     shc.construct_complex()
428     if not shc.break_routine:
429         if shc.disp:

```

File /opt/conda/envs/pycaret/lib/python3.8/site-packages/scipy/optimize/_shgo.py:75
1, in SHGO.construct_complex(self)

```

749         break
750         # Iterate complex, process minimisers
-> 751         self.iterate()
752         self.stopping_criteria()
754     # Build minimiser pool
755     # Final iteration only needed if pools weren't minimised every iteration

```

File /opt/conda/envs/pycaret/lib/python3.8/site-packages/scipy/optimize/_shgo.py:89
4, in SHGO.iterate(self)

```

893     def iterate(self):
-> 894         self.iterate_complex()
896         # Build minimizer pool
897         if self.minimize_every_iter:

```

File /opt/conda/envs/pycaret/lib/python3.8/site-packages/scipy/optimize/_shgo.py:93
3, in SHGO.iterate_delaunay(self)

```

927     def iterate_delaunay(self):
928         """
929         Build a complex of Delaunay triangulated points

```

```

930
931     Note: called with ``self.iterate_complex()`` after class initiation
932     """
--> 933     self.sampled_surface(infty_cons_sampl=self.infty_cons_sampl)
934     self.nc += self.n
935     self.n_sampled = self.nc

```

File /opt/conda/envs/pycaret/lib/python3.8/site-packages/scipy/optimize/_shgo.py:1264, in SHGO.sampled_surface(self, infty_cons_sampl)

```

1261 self.sorted_samples()
1263 # Find objective function references
-> 1264 self.fun_ref()
1266 self.n_sampled = self.nc

```

File /opt/conda/envs/pycaret/lib/python3.8/site-packages/scipy/optimize/_shgo.py:1382, in SHGO.fun_ref(self)

```

1379         break # Breaks the g loop
1381 if eval_f:
-> 1382     self.F[i] = self.func(self.C[i, :])
1383     self.fn += 1
1384 elif self.infty_cons_sampl:

```

File /opt/conda/envs/pycaret/lib/python3.8/site-packages/scipy/optimize/_optimize.py:539, in _wrap_scalar_function.<locals>.function_wrapper(x, *wrapper_args)

```

537 ncalls[0] += 1
538 # A copy of x is sent to the user function (gh13740)
--> 539 fx = function(np.copy(x), *(wrapper_args + args))
540 # Ideally, we'd like to have a true scalar returned from f(x). For
541 # backwards-compatibility, also allow np.array([1.3]), np.array([[1.3]]) et
c.
542 if not np.isscalar(fx):

```

File /opt/conda/envs/pycaret/lib/python3.8/site-packages/pycaret/classification/oo.py:2691, in ClassificationExperiment.optimize_threshold.<locals>.objective(x, *args)

```

2689 def objective(x, *args):
2690     probability_threshold = x[0]
-> 2691     model = self._create_model(
2692         estimator,
2693         verbose=False,
2694         system=False,
2695         probability_threshold=probability_threshold,
2696     )
2697     model_results = (
2698         self.pull(pop=True)
2699         .reset_index()
2700         (...)
2708         ]
2709     )
2710     model_results["probability_threshold"] = probability_threshold

```

File /opt/conda/envs/pycaret/lib/python3.8/site-packages/pycaret/internal/pycaret_experiment/supervised_experiment.py:1523, in _SupervisedExperiment._create_model(self, estimator, fold, round, cross_validation, predict, fit_kwargs, groups, refit, probability_threshold, experiment_custom_tags, verbose, system, add_to_model_list, X_train_data, y_train_data, metrics, display, model_only, return_train_score, **kwargs)

```

s)
1520         return model, model_fit_time
1521     return model
-> 1523 model, model_fit_time, model_results, _ = self._create_model_with_cv(
1524     model,
1525     data_X,
1526     data_y,
1527     fit_kwargs,
1528     round,
1529     cv,
1530     groups,
1531     metrics,
1532     refit,
1533     system,
1534     display,
1535     return_train_score=return_train_score,
1536 )
1538 # end runtime
1539 runtime_end = time.time()

```

File /opt/conda/envs/pycaret/lib/python3.8/site-packages/pycaret/internal/pycaret_experiment/supervised_experiment.py:1117, in _SupervisedExperiment._create_model_with_cv(self, model, data_X, data_y, fit_kwargs, round, cv, groups, metrics, refit, system, display, return_train_score)

```

1115 model_fit_start = time.time()
1116 with redirect_output(self.logger):
-> 1117     scores = cross_validate(
1118         pipeline_with_model,
1119         data_X,
1120         data_y,
1121         cv=cv,
1122         groups=groups,
1123         scoring=metrics_dict,
1124         fit_params=fit_kwargs,
1125         n_jobs=n_jobs,
1126         return_train_score=return_train_score,
1127         error_score=0,
1128     )
1129 model_fit_end = time.time()
1130 model_fit_time = np.array(model_fit_end - model_fit_start).round(2)

```

File /opt/conda/envs/pycaret/lib/python3.8/site-packages/sklearn/model_selection/_validation.py:266, in cross_validate(estimator, X, y, groups, scoring, cv, n_jobs, verbose, fit_params, pre_dispatch, return_train_score, return_estimator, error_score)

```

263 # We clone the estimator to make sure that all the folds are
264 # independent, and that it is pickle-able.
265 parallel = Parallel(n_jobs=n_jobs, verbose=verbose, pre_dispatch=pre_dispatch)
ch)
-> 266 results = parallel(
267     delayed(_fit_and_score)(
268         clone(estimator),
269         X,
270         y,
271         scorers,
272         train,

```



```

273         test,
274         verbose,
275         None,
276         fit_params,
277         return_train_score=return_train_score,
278         return_times=True,
279         return_estimator=return_estimator,
280         error_score=error_score,
281     )
282     for train, test in cv.split(X, y, groups)
283 )
284 _warn_or_raise_about_fit_failures(results, error_score)
285 # For callable scoring, the return type is only known after calling. If the
286 # return type is a dictionary, the error scores can now be inserted with
287 # the correct key.

```

File /opt/conda/envs/pycaret/lib/python3.8/site-packages/sklearn/utils/parallel.py:63, in Parallel.__call__(self, iterable)

```

58 config = get_config()
59 iterable_with_config = (
60     _with_config(delayed_func, config), args, kwargs)
61     for delayed_func, args, kwargs in iterable
62 )
--> 63 return super().__call__(iterable_with_config)

```

File /opt/conda/envs/pycaret/lib/python3.8/site-packages/joblib/parallel.py:1098, in Parallel.__call__(self, iterable)

```

1095     self._iterating = False
1097 with self._backend.retrieval_context():
-> 1098     self.retrieve()
1099 # Make sure that we get a last message telling us we are done
1100 elapsed_time = time.time() - self._start_time

```

File /opt/conda/envs/pycaret/lib/python3.8/site-packages/joblib/parallel.py:997, in Parallel.retrieve(self)

```

991 if (backend is not None and
992     hasattr(backend, 'abort_everything')):
993     # If the backend is managed externally we need to make sure
994     # to leave it in a working state to allow for future jobs
995     # scheduling.
996     ensure_ready = self._managed_backend
--> 997     backend.abort_everything(ensure_ready=ensure_ready)
998 raise

```

File /opt/conda/envs/pycaret/lib/python3.8/site-packages/joblib/_parallel_backend.py:586, in LokyBackend.abort_everything(self, ensure_ready)

```

583 def abort_everything(self, ensure_ready=True):
584     """Shutdown the workers and restart a new one with the same parameters
585     """
--> 586     self._workers.terminate(kill_workers=True)
587     self._workers = None
589     if ensure_ready:

```

File /opt/conda/envs/pycaret/lib/python3.8/site-packages/joblib/executor.py:74, in MemmappingExecutor.terminate(self, kill_workers)

```

73 def terminate(self, kill_workers=False):

```

```

---> 74     self.shutdown(kill_workers=kill_workers)
      75     if kill_workers:
      76         # When workers are killed in such a brutal manner, they cannot
      77         # execute the finalizer of their shared memmaps. The refcount of
      (...)
      83
      84         # unregister temporary resources from all contexts
      85         with self._submit_resize_lock:

```

File /opt/conda/envs/pycaret/lib/python3.8/site-packages/joblib/externals/loky/process_executor.py:1199, in ProcessPoolExecutor.shutdown(self, wait, kill_workers)

```

      1196         self._executor_manager_thread_wakeup.wakeup()
      1198     if executor_manager_thread is not None and wait:
-> 1199         executor_manager_thread.join()
      1201     # To reduce the risk of opening too many files, remove references to
      1202     # objects that use file descriptors.
      1203     self._executor_manager_thread = None

```

File /opt/conda/envs/pycaret/lib/python3.8/threading.py:1011, in Thread.join(self, timeout)

```

      1008         raise RuntimeError("cannot join current thread")
      1010     if timeout is None:
-> 1011         self._wait_for_tstate_lock()
      1012     else:
      1013         # the behavior of a negative timeout isn't documented, but
      1014         # historically .join(timeout=x) for x<0 has acted as if timeout=0
      1015         self._wait_for_tstate_lock(timeout=max(timeout, 0))

```

File /opt/conda/envs/pycaret/lib/python3.8/threading.py:1027, in Thread._wait_for_tstate_lock(self, block, timeout)

```

      1025     if lock is None: # already determined that the C code is done
      1026         assert self._is_stopped
-> 1027     elif lock.acquire(block, timeout):
      1028         lock.release()
      1029         self._stop()

```

KeyboardInterrupt:

```

In [11]: print('skipped last step, no results in over 30 minutes')
        print('continuing to the next segment of chapter 11')

```

skipped last step, no results in over 30 minutes
continuing to the next segment of chapter 11

```

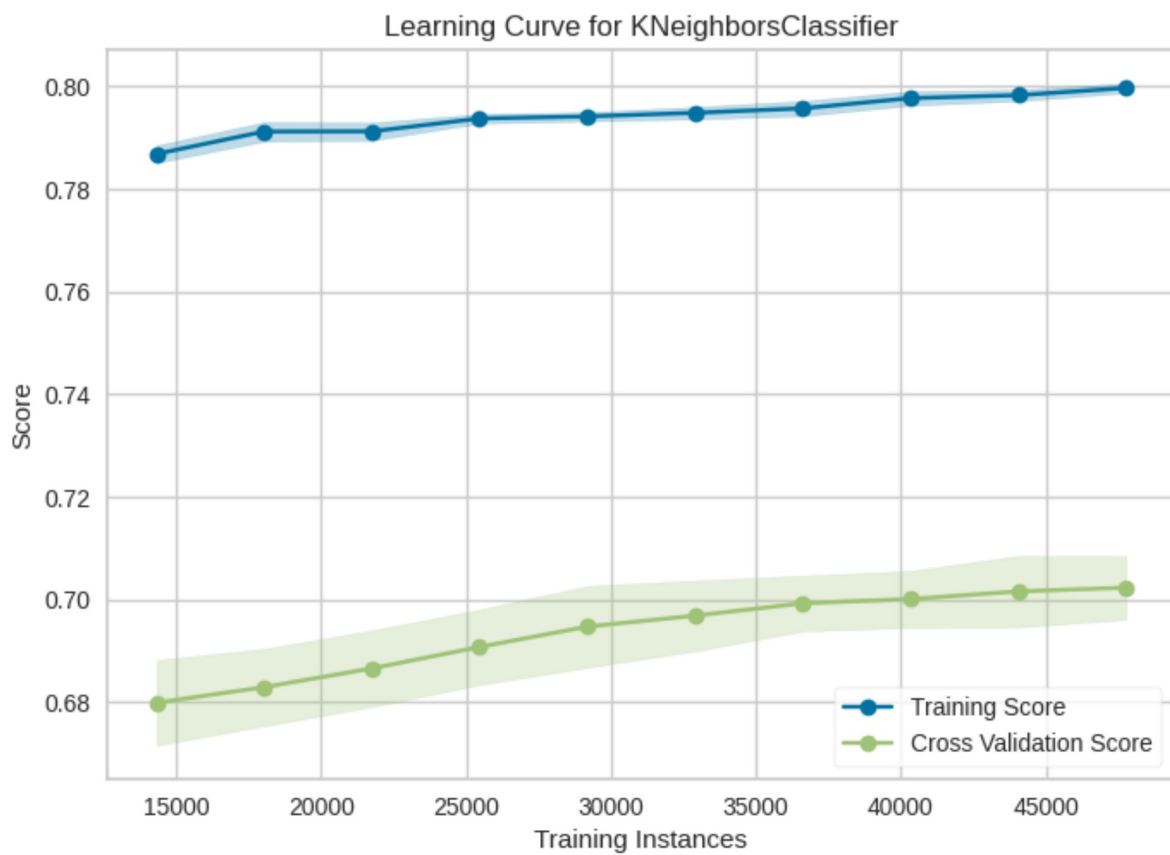
In [12]: from pandas import DataFrame
        import pycaret.classification as pcc
        final_best = pcc.finalize_model(besttuned)

```

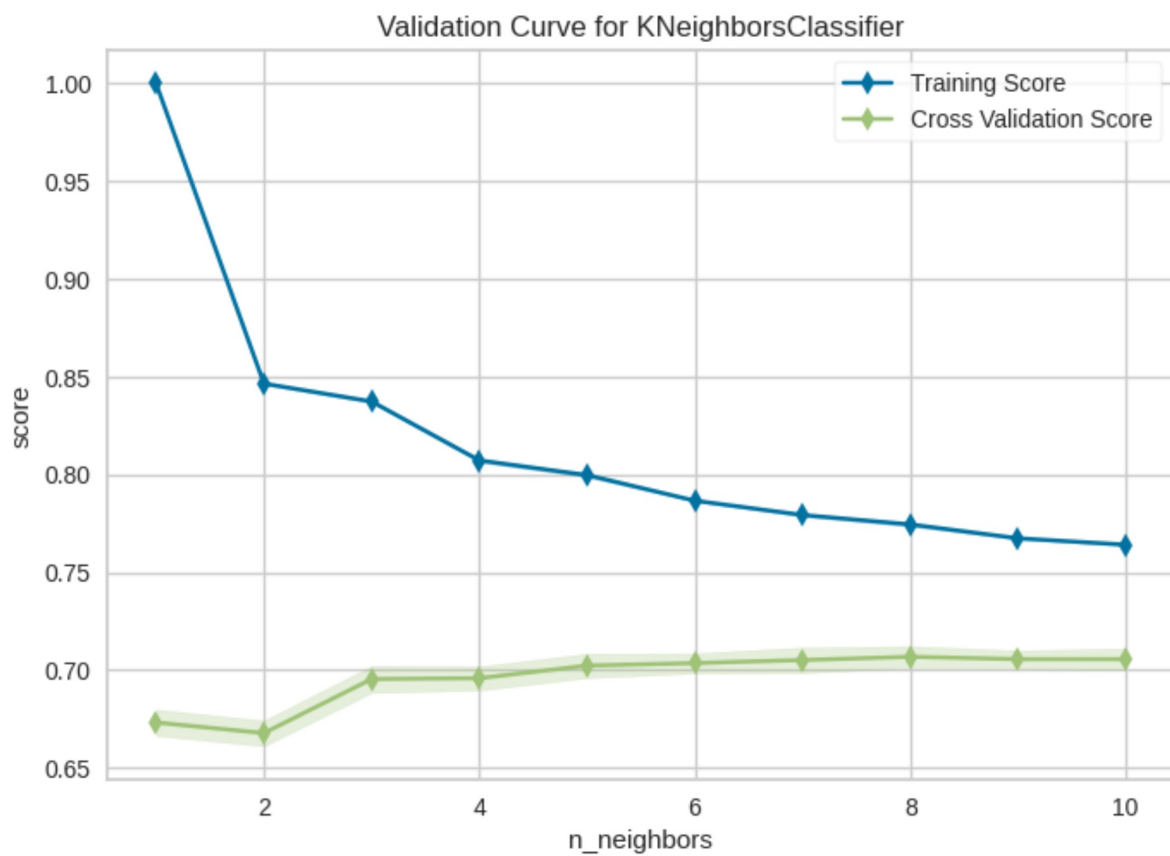
```

In [14]: data = DataFrame(['harry'], ['cerise'], ['gerlos'], ['ansna'], ['peytr'])
        data.columns=['FirstForename']
        print(data)

```

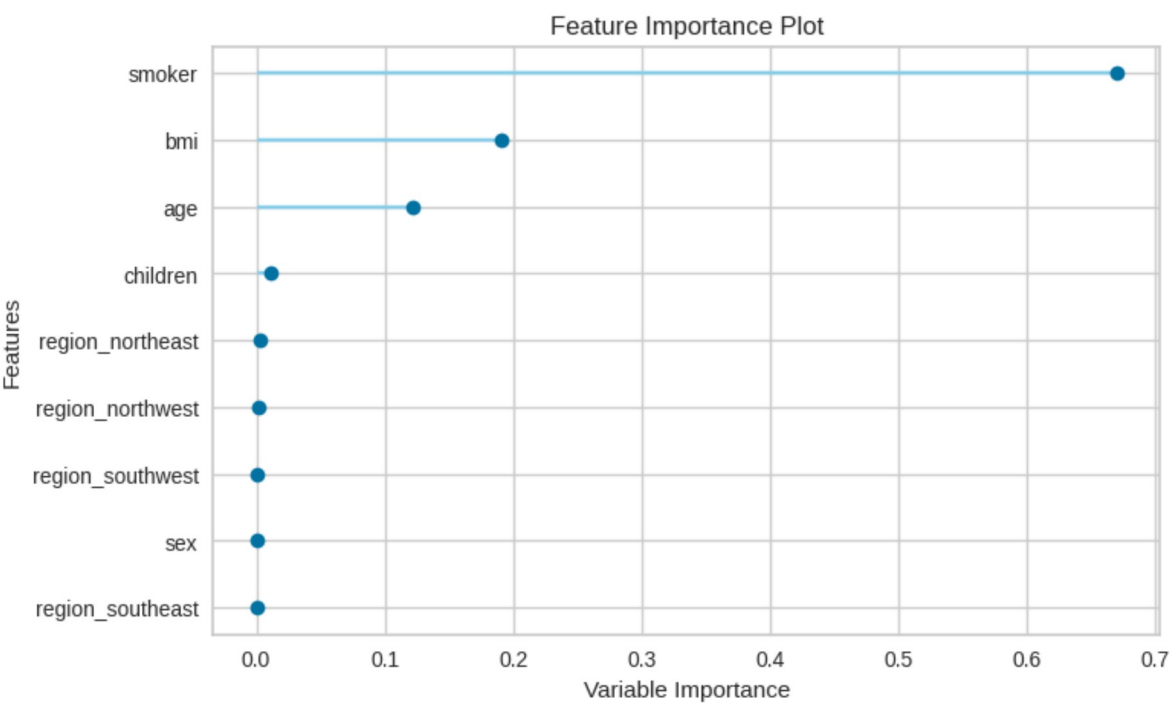
```
In [23]: plot_model(best, plot='vc')
```



```
In [24]: plot_model(best, plot='parameter')
```

Parameters	
algorithm	auto
leaf_size	30
metric	minkowski
metric_params	None
n_jobs	-1
n_neighbors	5
p	2
weights	uniform

```
In [26]: plot_model(best, plot='feature')
```



```
In [ ]:
```